USER INFORMATION
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AINIRON (Iron Sucrose Injection 20mg/mL)

Composition:
Each 5mL vial contains 100mg iron as iron sucrose (Iron (iii) hydroxide sucrose complex).

Product Description:
AINIRON is a solution for injection or concentrate for solution for infusion. The solution is dark brown, non transparent, aqueous solution.

Pharmacodynamics/Pharmacokinetics:
Iron sucrose is rapidly cleared from the plasma after intravenous injection with a terminal half-life of about 6 hours. A competitive exchange of iron takes place from the iron sucrose complex to the iron-binding protein transferrin. About 5% of a dose is eliminated via the kidneys in the first 4 hours after a dose.

Indication:
AINIRON is indicated for the treatment of iron deficiency anaemia in the following indications:-
- where there is a clinical need for a rapid iron supply,
- In patients who cannot tolerate oral iron therapy or who are non-compliant,
- In active inflammatory bowel disease where oral iron preparations are ineffective.

AINIRON should only be administered where the indication is confirmed by appropriate investigations (e.g. Hb, serum ferritin, serum iron).

Recommended Dosage:
Administration
AINIRON must only be administered by the intravenous route. This may be by a slow intravenous injection or by an intravenous drip infusion. AINIRON must not be used for intramuscular injection.
Before administration the first therapeutic dose, a test dose should be given. If any allergic reaction or intolerance occurs during administration, the therapy must be stop immediately.

Intravenous drip infusion
AINIRON must be diluted only in sterile 0.9% w/v sodium chloride solution:
- 100mg iron (5mL AINIRON) in maximum 100mL sterile 0.9% w/v sodium chloride solution
- 500mg iron (25mL AINIRON) in maximum 500mL sterile 0.9% w/v sodium chloride solution

For stability reasons, dilutions to lower AINIRON concentrations are not permissible.

As infusion, maximum tolerated single dose per day given not more than once per week:
- Patient above 70kg: 500mg iron (25mL AINIRON) in at least 3½ hours.
- Patient of 70kg and below: 7mg iron/kg body weight in at least 3½ hours.

Dilution must take place immediately prior to infusion and the solution should be administered as follows:
- 100 mg iron (5 mL AINIRON) in at least 15 minutes
- 200 mg iron (10 mL AINIRON) in at least 30 minutes
- 300 mg iron (15 mL AINIRON) in at least 1½ hours
- 400 mg iron (20 mL AINIRON) in at least 2½ hours

Before administration of the therapeutic dose in a new patient, the first 20mg iron in adults and in children with a body weight greater than 14kg and half the daily dose (1.5mg iron/kg) in children with body weight less than 14kg should be infused over 15 minutes as a test dose. If no adverse reactions occur, the remaining portion of the infusion can be administered at recommended speed.

Intravenous injection
AINIRON may be administered undiluted by slow intravenous injection as follows:
- 100 mg iron (5 mL AINIRON) in at least 5 minutes
- 200 mg iron (10 mL AINIRON) in at least 10 minutes

Before administration of the therapeutic dose in a new patient, a test dose of 1mL AINIRON (20mg iron) in adults and in children with a body weight greater than14kg and half the daily dose (1.5mg iron/kg) in children with body weight less than 14kg should be injected over 1 to 2 minutes. If no adverse reactions occur within a waiting period of 15 minutes, the remaining portion of the injection can be administered at recommended speed.

Injection into dialyser
AINIRON may be administered during a haemodialysis session directly into the venous limb of the dialyser under the same conditions as for intravenous injection.

Dosage
Calculation of dosage
The total cumulative dose of AINIRON, equivalent to the total iron deficit (mg), is determined by the haemoglobin level and body weight. The dose of AINIRON must be individually determined for each patient according to the total iron deficit calculated with the following formula:

Total iron deficit [mg] = body weight [kg] x (target Hb - actual Hb) [g/l] x 0.24* + depot iron [mg]

Below 35 kg body weight:
Target Hb = 130 g/L and depot iron = 15 mg/kg body weight

35 kg body weight and above:
Target Hb = 150 g/L and depot iron = 500 mg

*Factor 0.24 = 0.0034 x 0.07 x 1000

Iron content of haemoglobin ≅ 0.34%
Blood volume ≅ 7% of body weight
Conversion from g/L to mg/L = Factor 1000
Total amount of AINIRON to be administered (in mL) = $\frac{\text{Total iron deficit [mg]}}{\text{20mg/mL}}$

Body wieight [kg]	**Total number of vial AINIRON to be administered** (1 vial of AINIRON corresponds to 5mL)			
	Hb 60 g/L	Hb 75 g/L	Hb 90 g/L	Hb 105 g/L
5	1.5	1.5	1.5	1
10	3	3	2.5	2
15	5	4.5	3.5	3
20	6.5	5.5	5	4
25	8	7	6	5.5
30	9.5	8.5	7.5	6.5
35	12.5	11.5	10	9
40	13.5	12	11	9.5
45	15	13	11.5	10
50	16	14	12	10.5
55	17	15	13	11
60	18	16	13.5	11.5
65	19	16.5	14.5	12
70	20	17.5	15	12.5
75	21	18.5	16	13
80	22.5	19.5	16.5	13.5
85	23.5	20.5	17	14
90	24.5	21.5	18	14.5

To convert Hb (mM) to Hb (g/L), multiply the former by 16.1145.
If the total necessary dose exceeds the maximum allowed single dose, then the administration has to be split, please see section administration.

Calculation of dosage for iron replacement secondary to blood loss and to support autologous blood donation
The required AINIRON dose to compensate the iron deficit is calculated according the following formulas:
If the quantity of blood lost is known:
The administration of 200mg I.V. iron (=10mL AINIRON) results in an increase in haemoglobin which is equivalent to 1 unit blood (=400mL with 159g/L Hb content).
Iron to be replaced [mg] = number of blood units lost x 200 or
Amount of AINIRON needed needed [mL] = number of blood units lost X 10

If the Hb level is reduced:
Use the following formula considering that the depot iron does not need to be restored.
Iron to be replaced [mg] = body weight [kg] x 0.24 x (target Hb-actual Hb) [g/L]
e.g.: body weight 60kg, Hb deficit = 10g/L
- ≅ 150mg iron to be replaced
- 7.5mL AINIRON needed

Normal posology
Adults and the elderly: 5-10 mL AINIRON (100-200 mg iron) one to three times a week depending on the haemoglobin level.
Children: There is limited data on children under study conditions. If there is a clinical need, it is recommended not exceed 0.15 mL AINIRON (3 mg iron) per body weight one to three times per week depending on the haemoglobin level.

Maximum tolerated dose
As injection, maximum tolerated dose per day, given not more than three times per week:
• 200 mg iron (10 ml AINIRON) injected over at least 10 minutes.

As infusion, maximum tolerated single dose per day given not more than once per week:
• Patient above 70kg: 500mg iron (25mL AINIRON) in at least 3½ hours.
• Patient of 70kg and below: 7mg iron/kg body weight in at least 3½ hours
The maximum tolerated single dose is 7 mg iron per kg body weight given once per week, but not exceeding 500mg iron. Administration time and dilution see section administration. The infusion times given in section administration must be strictly adhered to, even if the patient does not receive the maximum tolerated single dose.

Route of Administration: INTRAVENOUS (I.V.) ONLY

Contraindications:
The use of AINIRON is contraindicated in cases of:
• anaemia not caused by iron deficiency
• iron overload or disturbances in utilisation of iron
• known hypersensitivity to AINIRON or any of its inactive component
• pregnancy first trimester

Warnings and Precautions:
Parenterally administered iron preparations can cause allergic or anaphylactoid reactions, which may be potentially fatal. Therefore, antiallergic treatment should be place with established cardio-pulmonary resuscitation procedures.
In patients with history of asthma, eczema, other atopic allergies or allergic reactions to other parenteral iron preparations, AINIRON should be administered with care as they are particularly at risk of an allergic reaction. However, it was shown in a study with limited number of iron dextran sensitive patients that AINIRON could be administered with no complications.
AINIRON should be administered with care in patients with liver dysfunction.
AINIRON must be used with care in patients with acute or chronic infection who have excessive ferritin values as parenterally administered iron can unfavourably influence a bacterial or viral infection.
Hypotensive episodes may occur if the injection is administered too rapidly.
Paravenous leakage must be avoided because leakage of AINIRON at the injection site may lead to pain, inflammation, tissue necrosis and brown discoloration of the skin

Caution:
AINIRON must be used immediately after dilution. Single dose container. Discard all unused contents. Do not use if the solution is discoloured and leakage is detected.

Interactions with Other Medicaments:
As with all parenteral iron preparations, AINIRON should not be administered concomitantly with oral iron preparations since the absorption of oral iron is reduced. Therefore an oral iron therapy should at least be started 5 days after the last injection of AINIRON.

Statement on Usage During Pregnancy and Lactation:
Data on a limited number of exposed pregnancies indicated no adverse effects of AINIRON on pregnancy or on the health of the foetus/newborn child. No well-controlled studies in pregnant women are available to date. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development.
Nevertheless, risk/benefit evaluation is required.
Non metabolised AINIRON is unlikely to pass into the mother's milk. No well-controlled clinical studies are available to date. Animal studies do not indicate direct or indirect harmful effects to nursing child.

Adverse Effects / Undesirable Effects:
The most frequently reported adverse drug reactions (ADRs) of iron sucrose were transient taste perversion, hypotension, fever and shivering, injection site reactions and nausea. Non-serious anaphylactoid reactions occurred rarely.
In general anaphylactoid reactions are potentially the most serious adverse reactions (see "Warnings and Precautions").
The following adverse drug reactions have been reported in temporal relationship with the administration of iron sucrose, with at least a possible causal relationship:

Nervous system disorders
Common : transient taste perversions (in particular metallic taste).
Uncommon: headache, dizziness.
Rare : paraesthesia, syncope, loss of consciousness, burning sensation.

Cardio-vascular disorders
Uncommon: hypotension and collapse, tachycardia and palpitations.
Rare : hypertension.

Respiratory, thoracic and mediastinal disorders
Uncommon: bronchospasm, dyspnoea.

Gastrointestinal disorders
Uncommon: nausea; vomiting, abdominal pain, diarrhoea.

Skin and subcutaneous tissue disorders
Uncommon: pruritus, urticaria, rash, exanthema, erythema.

Musculoskeletal, connective tissue and bone disorders
Uncommon: muscle cramps, myalgia.

General disorders and administration site disorders
Uncommon: fever, shivering, flushing, chest pain and tightness. Injection site disorders such as superficial phlebitis, burning, swelling.
Rare : arthralgia, peripheral oedema, fatigue, asthenia, malaise, feeling hot, oedema.

Immune system disorders
Rare : anaphylactoid reactions.
Moreover, in spontaneous reports the following adverse reactions have been reported:
Isolated cases: reduced level of consciousness, light-headed feeling, confusion, angio-oedema, and swelling of joints, hyperhidrosis, back pain, bradycardia, chromaturia.

Overdose and Treatment:
Overdosage can cause acute iron overloading which may manifest itself as haemosiderosis. Overdosage should be treated, if required, with an iron chelating agent.

Storage Conditions:
Do not store above 30°C. Store in original carton. Do not freeze.

Shelf life:
3 years from manufacturing date in the proposed storage condition. The solution should be used immediately after dilution. Do not use after expiry date.

Dosage form and packaging available:
5mL X 20 vials per box

Manufacturer / Product Registration Holder :
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